

BGDDR User Manual

Bangladesh Glomerular Disease and Diabetes Registry
BIRDEM General Hospital - Department of Nephrology

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Application: BGDDR single-user Windows desktop app

This manual explains how to use BGDDR in day-to-day clinical work and registry operations: starting the app, logging in, registering patients, recording baseline and follow-up data, writing prescriptions, entering labs and biopsies, managing consent, enrolling patients into studies, reviewing analytics, exporting data, and performing administrator tasks.

For installation and packaging details, see `DESKTOP_DEPLOYMENT.md`. This manual assumes the app is already installed or running from `BGDDR.exe`.

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1. Purpose of BGDDR

BGDDR is both a clinical workflow application and a research registry. The app is designed around one practical idea:

The prescription is the visit.

At each patient encounter, the clinician records the visit, writes the current full prescription, orders or records investigations, and schedules the next review. When the prescription is finalized, BGDDR also updates the research-grade medication exposure table automatically. This means routine clinical work creates structured longitudinal research data without duplicate entry.

BGDDR supports:

- Patient registration and longitudinal GN/diabetes follow-up.
 - Baseline demographic, clinical, comorbidity, presentation, examination, and serology capture.
 - Clinical encounters with response status, disease phase, treatment adjustment, advice, and next due date.
 - Laboratory ordering and longitudinal results, including automatic eGFR derivation from serum creatinine.
 - Biopsy records, GN diagnosis, Oxford MEST-C, lupus class, FSGS variant, and membranous nephropathy features.
 - Bilingual prescription printing with safety checks and immutable finalization.
 - Medication exposure reconstruction from prescriptions.
 - Consent versioning and withdrawal.
 - Registry-embedded observational studies and randomized trials.
 - Safety events, infection incidence, and study safety summaries.
 - Outcomes, survival analysis, Cox regression, eGFR slope, competing risks, and cohort summaries.
 - De-identified and identified exports to CSV, Excel, and SPSS.
 - Audit trail, user roles, backup, and restore.
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2. First Launch and Login

Start the Desktop App

1. Open the BGDDR folder.
2. Double-click `BGDDR.exe`.
3. A small status window appears saying the registry is running.
4. Your browser opens at `http://127.0.0.1:8000/`.

Keep the status window open while using the app. Use its Stop button to shut down the local server cleanly.

First Administrator Account

On the first launch, the app asks you to create the first administrator account. This account is the main superuser. It can access all modules, the Django admin site, backup/restore tools, reference data, and user management.

Recommended practice:

- Use a named account rather than a shared password where possible.
- Store the first administrator password securely.
- Create separate accounts for clinicians, coordinators, statisticians, and other staff.

Login

1. Go to `http://127.0.0.1:8000/login/`.
2. Enter username and password.
3. Click Sign in.

After login, the dashboard opens. Use Logout from the top-right account menu when finished.

Password Reset

Users can request password reset from `/password-reset/` if email/reset settings are configured. An administrator can also reset passwords from:

- Admin -> Users -> select user -> change password
- or from a shell: `python manage.py changepassword <username>`

3. Roles and Access

BGDDR uses role-based access control. Roles are stored as Django groups and are assigned through user profiles or invitations.

Role	Typical user	Main purpose
Data Manager	Registry lead, admin	Full data management; can perform identified exports
Coordinator	Study/clinic coordinator	Patient registration, visits, labs, consent, enrolment
Investigator	Clinician, PI	Clinical entry, prescriptions, outcomes review
Pathologist	Renal pathologist	Biopsy and central pathology review
Statistician	Analyst	Analytics and de-identified exports
Read-only	Auditor, observer	View-only access
Superuser	System administrator	Full access including admin and maintenance

The first account created by BGDDR.exe is a superuser. Superusers bypass normal role restrictions, so they should be limited to trusted administrators.

4. Navigation Overview

The app uses a patient-centered workflow. The sidebar and patient pages are the main navigation surfaces.

Sidebar Sections

Section	Use
Dashboard	Registry overview, quick links, counts, and worklist summary
Patients	Search, register, edit, and open patient records
Worklist	Due visits, overdue visits, and clinic roster
Prescriptions	Recent prescriptions and prescription follow-up
Analytics	Survival, Cox, eGFR slope, competing risks, and cohort tools
Studies	Study portfolio, recruitment funnel, and study dashboards
Safety	Adverse event summaries and infection incidence
Pathology	Biopsy review, discordant cases, and agreement statistics
Biomarkers	Patient biomarker trends and PLA2R predictor
Export	Research dataset downloads
Admin	Back-office data and system administration

Header Tools

Use the header to:

- Return to the dashboard.
- Jump quickly to a patient by ID or name.

- Open the user profile.
 - Log out.
-

5. Daily Clinic Workflow

The recommended daily workflow is:

1. Open Worklist and identify due or overdue patients.
2. Open the patient record.
3. Update patient demographics if needed.
4. Record the follow-up visit.
5. Order investigations or enter available lab results.
6. Record adverse events, admission, relapse, consent, or study enrolment if relevant.
7. Write the full current prescription.
8. Review safety warnings.
9. Finalize and print or download the prescription.
10. Confirm the next due date.

Important rule for prescriptions:

Each prescription should contain the patient's complete current regimen, not only the changed medicines. The reconciliation engine compares the finalized prescription against previous open medication episodes.

6. Patient Registration

Open:

- `Patients -> Add patient`
- or `/patients/add/`

Required and Common Fields

Field	Notes
Name	Patient's name
Hospital ID	Hospital/BIRDEM registration number, optional but recommended
Phone	Useful for duplicate detection and follow-up
Sex	Male, Female, or Other
Date of birth	Used for age and eGFR derivation
Enrollment date	Registry entry date
Cohort	Patient category
Diabetes status	No diabetes, Type 1, Type 2, or other/secondary
Primary diagnosis	Curated GN diagnosis list

The patient ID is generated automatically in the format BGD-00001 , BGD-00002 , and so on. Do not type this manually.

Duplicate Check

As patient details are entered, BGDDR can detect likely duplicates using name, phone, hospital ID, and other matching fields. Review possible matches before saving a new patient.

After Saving

After saving, BGDDR opens the patient detail page. From there, you can add:

- Baseline assessment.
 - Follow-up encounter.
 - Lab order.
 - Lab results.
 - Biopsy.
 - Prescription.
 - Consent.
 - Study enrolment.
 - Adverse event.
 - Admission.
 - Relapse.
 - Treatment exposure.
-

7. Patient Detail Page

The patient detail page is the main working screen for a single patient.

It displays:

- Patient identity and demographics.
- Registration status and disease phase.
- Latest renal function where available.
- Baseline summary.
- Encounters and visit history.
- Lab trends and result tables.
- Prescriptions.
- Treatment exposures.
- Biopsies and pathology.
- Study enrolments.
- Consents.
- Outcomes.
- Safety events.

Common actions:

Button/action	Use
Edit	Correct demographic or core patient details
Baseline	Add or update baseline assessment
Follow-up	Record a visit
Lab order	Order tests for a visit
Enter results	Enter lab/serology results
Biopsy	Add biopsy and pathology details
Prescription	Write a prescription for an encounter
Consent	Record or withdraw consent
Enrol in study	Screen/enrol patient into a study
Treatment	Add external or prior medication exposure
Adverse event	Record safety event
Admission	Record inpatient work-up
Relapse	Record relapse episode
Recompute outcomes	Refresh derived outcome row

8. Baseline Assessment

Open:

- Patient page -> Baseline
- or `/patients/<id>/baseline/`

Baseline is the enrolment-time clinical snapshot. It should be completed as early as possible because it feeds outcomes, analyses, and exports.

Sections Captured

Section	Examples
Social/demographic	Residence division, education, occupation, income
Medical history	Hypertension, CVD, previous kidney disease, autoimmune disease
Diabetes burden	Duration, HbA1c, retinopathy, neuropathy, diabetic foot
Anthropometry/vitals	Height, weight, BMI, BP, pulse, temperature
Presentation	Syndrome, symptoms, oedema, urinary sediment
Examination	Volume status, skin, joint, fundoscopy
Point-of-care labs	Creatinine, proteinuria, albumin, Hb, potassium
Serology	ANA, anti-dsDNA, ANCA, anti-PLA2R, anti-GBM, ASO, C3, C4, Gd-IgA1

Automatic Calculations

BGDDR automatically:

- Calculates BMI from height and weight.
- Classifies BMI using Asian cutoffs.
- Converts creatinine from umol/L to mg/dL if that unit is selected.
- Derives eGFR when creatinine is entered.
- Stores lab values as longitudinal LabResult rows.

Good Practice

- Enter dates accurately.
- Use dropdowns rather than free text where available.
- Enter creatinine with the correct unit.
- Record both numeric serology values and qualitative reads when available.
- Use notes only for information that does not fit structured fields.

9. Follow-up Visits

Open:

- Patient page -> Follow-up
- or `/patients/<id>/followup/`

Each clinic visit should be recorded as a Clinical Encounter.

Main Fields

Field	Use
Encounter date	Date patient was seen
Encounter type	Baseline/enrollment, scheduled follow-up, unscheduled/sick visit
Seen by	Clinician responsible for the encounter
Clinic location	Clinic/site/room, if needed
BP and weight	Longitudinal clinical measures
Oedema grade	0 none to 4 anasarca
Symptoms	Frothy urine, haematuria, breathlessness, etc.
Clinician response	Clinical assessment at the visit
Disease phase	Current phase; can be left blank if workflow sets it
Treatment adjusted	Tick if therapy changed
Advice	Prints on prescription; Bangla text is supported
Next due date	Drives dashboard and worklist

After Saving

After a follow-up is saved, you can:

- Write a prescription linked to that visit.
- Order labs.
- Enter lab results.
- Record adverse events.
- Record relapse or admission if relevant.

10. Laboratory Orders and Results

BGDDR separates ordering tests from receiving results.

Lab Order

Open:

- Patient page -> Lab order
- or `/patients/<id>/lab-order/`

Use this when tests are requested at a visit.

You can:

- Choose a predefined lab panel.
- Add individual tests.
- Add notes such as urgency or special instructions.

The order links to the encounter and patient. Its status is updated as results are recorded.

Lab Results

Open:

- Patient page -> Enter results
- or `/patients/<id>/lab-results/`

Use this when results are available, even if they were not ordered inside BGDDR.

Result groups include:

- Kidney function and urine.
- Immunology/serology.
- Infection screen.
- Metabolic tests.
- Other active tests from the catalog.

Creatinine and eGFR

Creatinine may be entered in:

- mg/dL
- umol/L

When creatinine is saved, BGDDR stores it canonically in mg/dL and derives eGFR using CKD-EPI 2021. The latest eGFR is cached on the patient record and is used in prescription renal-safety checks.

Proteinuria

Proteinuria values feed remission and outcome logic. Use the most reliable available measure:

- 24-hour urine total protein when available.
- UPCR as fallback.
- UACR where relevant.

Qualitative Results

Some tests use qualitative values such as:

- Positive.
- Negative.
- Equivocal.
- Low.
- Normal.
- High.

For serology, enter both the numeric value and the qualitative read when both are available.

11. Biopsy and Pathology

Open:

- Patient page -> Biopsy
- or `/patients/<id>/biopsy/`

Core Biopsy Data

Record:

- Biopsy date.
- Adequacy.
- Number of glomeruli.
- Global sclerosis.
- IFTA.
- Crescents or necrosis.
- Vascular lesions.
- IF pattern.
- EM findings.
- Notes.

GN Diagnosis

Enter the structured GN diagnosis and broad group. This drives disease-specific analyses and export variables.

Disease-Specific Scoring

Where applicable, complete:

- IgA nephropathy: Oxford MEST-C.

- Lupus nephritis: ISN/RPS class and activity/chronicity fields.
- FSGS: variant.
- Membranous nephropathy: stage/features.

Pathology Review

Open:

- Pathology -> Biopsy review
- /pathology/biopsy/<id>/review/
- /pathology/discordant/
- /pathology/agreement/

The pathology section supports local review, central review, adjudication, discordance tracking, and agreement statistics.

12. Prescriptions

Open:

- Patient page -> Prescription
- or /patients/<id>/prescription/

Prescription is the most important daily workflow. It produces the printable clinical prescription and updates the research exposure history.

Before Writing a Prescription

Confirm that:

- The correct patient is open.
- A follow-up encounter exists for the visit date.
- Recent labs are entered if they affect safety decisions.
- Current medications are known.

Prescription Fields

Field	Use
Diagnosis	Printed diagnosis text
Comorbidities	Relevant comorbidities printed on the slip
Drug rows	Full current regimen
Investigations advised	Tests ordered or advised
Advice	General advice; Bangla supported
Stop notes	Reasons for stopped medications

Medication Rows

For each medication, enter:

- Drug.
- Brand, if needed.
- Strength.
- Dose.
- Route.
- Frequency.
- Timing.
- Duration.
- Bangla instruction.

Leave unused rows blank.

Always Enter the Full Current Regimen

BGDDR compares the finalized prescription with the patient's open medication episodes.

What happens in prescription	What BGDDR does
New drug appears	Opens a new exposure episode
Same drug and same regimen continues	Keeps episode open
Dose/frequency/route changes	Closes old episode and opens a new one
Previously open drug is missing	Closes the episode

If you only enter changed medicines, BGDDR will think the omitted medicines were stopped. Therefore, every finalized prescription should list all medicines the patient should continue taking.

Draft, Preview, and Finalize

1. Save the prescription as a draft.
2. Open Preview.

3. Review diagnosis, medicine rows, advice, investigations, and patient details.
4. Check safety warnings.
5. Finalize only when correct.

Finalization:

- Marks the prescription as final.
- Generates a stable content hash.
- Prevents silent editing of the finalized clinical document.
- Reconciles treatment exposure episodes.
- Enables printing or PDF download.

Printing and PDF

Use:

- `/prescriptions/<id>/preview/`
- `/prescriptions/<id>/pdf/`
- `/prescriptions/<id>/html/`

The PDF route uses the app's PDF renderer. If PDF dependencies are unavailable, the HTML version can still be printed from the browser.

Common Prescription Mistakes

Mistake	Consequence	Prevention
Entering only changed drugs	Omitted drugs may be closed as stopped	Enter full current regimen
Wrong creatinine unit in labs	eGFR and safety warning may be wrong	Check unit before saving labs
Finalizing too early	Final document becomes immutable	Use Preview carefully
Missing next due date	Patient may not appear correctly on worklist	Set next due date in encounter
Using free text instead of formulary drug	Exposure classification may be incomplete	Choose from DrugMaster dropdown

13. Treatment Exposure History

Open:

- Patient page -> Treatment
- or `/patients/<id>/treatment/`

Treatment exposures are research-grade medication episodes. Most in-clinic exposures are created automatically by prescription reconciliation.

Use manual Treatment entry for:

- Prior medications before registry entry.
- External prescriptions from another clinic.
- Medications known from records but not prescribed in BGDDR.

Manual exposure fields include:

- Drug.
- Dose.
- Frequency.
- Route.
- Start date.
- Stop date.
- Ongoing status.
- Stop reason.

There should be only one ongoing episode per patient per drug. Close an old episode before adding a new ongoing episode for the same drug.

14. Consent

Open:

- Patient page -> Consent
- or `/patients/<id>/consent/`

BGDDR stores consent as a versioned, withdrawable record.

Consent Types

Type	Meaning
Registry	Participation in the registry
Biobank	Sample storage
Genetic	Genetic testing
Imaging	Digital pathology/imaging
Trial	Registry-embedded trial participation

Recording Consent

1. Choose consent type.
2. Select ICF version.
3. Enter consent date, or leave blank to default to today.

4. Add scope or notes if useful.
5. Save.

When a new consent of the same type is granted, it supersedes the previous current version. The version chain is retained.

Withdrawing Consent

Use the Withdraw action beside the current consent. Withdrawal preserves the history and marks the consent inactive.

Trial Consent Gate

For randomized trials or studies configured to require trial consent, BGDDR will not enrol the patient until active Trial consent exists.

15. Admissions and Relapse

Admission

Open:

- Patient page -> Admission
- or `/patients/<id>/admission/`

Use Admission for inpatient work-up episodes. Capture:

- Admit date.
- Discharge date.
- Ward.
- Reason.
- Related biopsy.
- Whether baseline was captured before discharge.
- Discharge advice.

Relapse

Open:

- Patient page -> Relapse
- or `/patients/<id>/relapse/`

Use Relapse to document flare/relapse episodes. Capture:

- Relapse date.
- Relapse type.

- Criteria.
- Action taken.

Recording a relapse also supports outcome and disease-phase tracking.

16. Adverse Events and Safety

Open:

- Patient page -> Adverse event
- or `/patients/<id>/adverse-event/`
- Sidebar -> Safety

Adverse Event Fields

Field	Use
Onset date	Date event began
Category	Infection, steroid toxicity, haematologic, hepatic, nephrotoxicity, etc.
Infection type	Required when category is infection
Description	Short clinical description
Severity	Mild to fatal
Hospitalization	Marks event serious
Outcome	Recovered, recovering, ongoing, fatal, etc.
Suspected drug	Supports infection/safety by agent analysis
Relatedness	Unrelated to definite
Encounter	Optional linked visit
Notes	Additional details

BGDDR automatically marks an event serious if:

- Hospitalization is ticked.
- Severity is life-threatening.
- Severity is fatal.

Safety Pages

Page	Use
<code>/safety/summary/</code>	Overall adverse-event summary
<code>/safety/infection-incidence/</code>	Infection incidence analysis
<code>/safety/study/<code>/</code>	Study-specific safety summary

17. Scheduling and Worklist

Scheduling is driven mainly by encounter `next_due_date`.

Open:

- Sidebar -> Worklist
- `/clinic/worklist/`
- `/scheduling/due/`
- `/scheduling/overdue/`
- `/scheduling/roster/`

Use Worklist to:

- Find patients due today or soon.
- Identify overdue patients.
- Review clinic roster.
- Prioritize follow-up calls.

Good practice:

- Enter a next due date at every follow-up.
 - Update the date if the appointment changes.
 - Use the worklist at the beginning of clinic.
-

18. Studies and Randomization

Open:

- Sidebar -> Studies
- Patient page -> Enrol in study
- `/clinic/studies/`
- `/patients/<id>/enroll/`

- `/studies/<code>/dashboard/`

Study Types

Type	Behavior
Observational	Enrols eligible patients, no randomization
Quasi-experimental	Enrols without random allocation
Randomized controlled trial	Requires arms and randomization scheme

Enrolling a Patient

1. Open patient.
2. Click Enrol in study.
3. Choose study.
4. Set screened/enrolled dates if needed.
5. Save.

BGDDR then:

- Screens eligibility where coded.
- Checks trial consent if required.
- Randomizes if the study is randomized.
- Records arm, stratum, sequence position, user, and timestamp.

Study Statuses

Status	Meaning
Planning	Defined but not open
Recruiting	Open for enrolment
Active	Follow-up ongoing, closed to recruitment
Closed	Study closed

Only studies not closed are listed in the enrolment form.

Preloaded Study Portfolio

BGDDR includes a preloaded GN/diabetes research portfolio. Examples include:

- BGDDR-REGISTRY.
- ADVANCED-DKD-IGAN.
- HCQ-IGAN-ADVANCED.
- MMF-IGAN.
- MEST-C-PREDICTOR.

- RTX-MN-DOSE.
- LN-IMPLEMENTATION.
- IGAN-BUDESONIDE.
- MN-RTX-CNI.
- FSGS-STEROID-CNI.
- SGLT2-GN.
- FINERENONE-GN.

The administrator can refresh these with:

```
python manage.py seed_studies
```

19. Outcomes and Analytics

BGDDR computes outcomes from baseline, lab, event, treatment, and follow-up data.

Open:

- Sidebar -> Analytics
- Patient page -> Recompute outcomes
- `/clinic/analytics/`

Patient Outcome

Endpoint:

```
/analytics/patient/<patient_id>/outcome/
```

Use this to inspect one patient's computed outcome row. Add `?recompute=1` to refresh.

Cohort Analytics Endpoints

Endpoint	Purpose
<code>/analytics/cohort/survival/</code>	Kaplan-Meier survival and log-rank
<code>/analytics/cohort/survival/plot/</code>	SVG survival plot
<code>/analytics/cohort/cox/</code>	Cox proportional hazards model
<code>/analytics/cohort/egfr-slope/</code>	eGFR slope mixed model
<code>/analytics/cohort/cif/</code>	Competing-risk cumulative incidence
<code>/analytics/cohort/summary/</code>	Cohort summary table

Common Grouping Options

- Diabetes.

- Diagnosis.
- Cohort.
- Drug class, such as `drug:sglt2i` or `drug:hcq`.
- Study arm, such as `study:MN-RTX-CNI`.

Common Endpoints

- Composite kidney event.
- ESKD.
- Death.
- Sustained 40% eGFR decline.
- Sustained 50% eGFR decline.
- Complete remission.
- Partial remission.
- Any remission.
- IgAN proteinuria response.

Recomputing Outcomes

Use Recompute outcomes after major data corrections such as:

- Backfilled labs.
- Added hard endpoint events.
- Corrected baseline values.
- Corrected diagnosis.

20. Data Export

Open:

- Sidebar -> Export
- `/clinic/export/`
- `/exports/research-dataset/`
- `/exports/data-dictionary/`

BGDDR exports a one-row-per-patient research dataset.

Export Formats

Format	Use
CSV	Plain analysis-ready data
Excel	Workbook with dataset and data dictionary
SPSS SAV	Labelled SPSS file with variable/value labels

De-identified by Default

The standard export excludes direct identifiers. It is suitable for analysis and sharing within approved research workflows.

Identified Export

Identified export includes fields such as name/phone/hospital identifiers and is restricted to Data Manager or equivalent privileged users.

Command-Line Export

From the app folder:

```
python manage.py export_dataset
python manage.py export_dataset --format xlsx
python manage.py export_dataset --format sav
python manage.py export_dataset --identified --format xlsx
```

Command-line exports are written into the `Exports/` folder.

21. User and Role Administration

Open:

- Sidebar -> Admin
- `/users/`
- `/users/invite/`
- `/admin/`

Invite a User

Recommended workflow:

1. Open Users -> Invite user.
2. Enter email.
3. Select role.
4. Save.

5. Send the invitation link to the user.
6. User opens the link and sets a password.

Manage Existing Users

Administrators can:

- View user list.
- Change role.
- Update profile.
- Reset password.
- Disable accounts by removing active status in the admin.

Profile

Users can open `/profile/` to edit:

- Department.
 - Phone.
-

22. Reference Data Administration

Open:

- `/admin/`

The admin site is used for reference data and back-office maintenance.

Drug Formulary

Admin model:

- Treatments -> Drug masters.

Fields include:

- Generic name.
- Brand names.
- Drug class.
- Available strengths.
- Default frequency.
- Default route.
- Available routes.
- Strengths by route.
- Renal dose adjustment flag.
- eGFR caution threshold.

- Active status.

Drug class is important because analytics uses it to classify exposures.

Lab Tests

Admin model:

- Labs -> Lab tests.

Fields include:

- Code.
- Name.
- LOINC.
- Default unit.
- Value type.
- Reference range.
- Derived flag.
- Active status.

Do not enter eGFR as a manual test if it is configured as derived. Creatinine drives eGFR calculation.

Lab Panels

Admin model:

- Labs -> Lab panels.

Panels allow one-click lab ordering.

Advice Templates

Admin model:

- Prescriptions -> Advice templates.

Use these for reusable prescription advice snippets such as steroid counseling, sick-day rules, nephrotic diet, and infection precautions.

Studies

Admin models:

- Studies -> Studies.
- Studies -> Study arms.
- Studies -> Study enrollments.

For randomized trials, define:

- Study code.

- Title.
 - Status.
 - Primary endpoint.
 - Randomization scheme.
 - Block multipliers.
 - Stratification variables.
 - Random seed.
 - Trial consent requirement.
 - Arms and allocation ratios.
-

23. Backup and Restore

BGDDR is designed for single-user SQLite desktop operation. Data lives beside the executable unless `BGDDR_DATA_DIR` is set.

Automatic Backups

The desktop launcher takes:

- One backup at startup.
- Periodic backups while running, default every 6 hours.

Backups are stored in:

`Backups/`

The newest backups are retained according to the configured retention limit.

Manual Backup

From the app folder:

```
python manage.py backup_db
python manage.py backup_db --list
```

Restore

Stop BGDDR before restoring.

```
python manage.py restore_db --list
python manage.py restore_db <backup-file-name> --yes
```

Before restore, BGDDR creates a pre-restore safety snapshot so the restore can be reversed if needed.

OneDrive Use

If the BGDDR folder is inside OneDrive, data and backups can sync. However:

- Open the app in only one computer at a time.
 - Wait for sync to complete before opening on another computer.
 - Do not edit or replace `db.sqlite3` while BGDDR is running.
-

24. Data Quality and Audit Trail

Audit Trail

BGDDR records changes with:

- Model/object.
- Action.
- Field name.
- Old value.
- New value.
- User.
- Timestamp.
- Change reason where provided.

API writes can also carry a change reason using the `x-Change-Reason` header.

Data Quality Checks

Use the Quality page and admin review to look for:

- Duplicate patients.
- Missing baseline assessments.
- Missing next due dates.
- Missing diagnosis.
- Missing consent for trial participants.
- Patients with labs but no baseline.
- Unfinalized prescriptions.
- Open medication episodes that do not match current clinical reality.

Data Entry Principles

- Prefer dropdown values over free text.
- Enter dates carefully.
- Use correct lab units.
- Finalize prescriptions only after review.

- Keep identifiers accurate.
 - Record reasons for stopping important therapies.
 - Recompute outcomes after major backfilled data entry.
-

25. Troubleshooting

App Does Not Open

Check:

- Is `BGDDR.exe` blocked by Windows security?
- Is the `_internal` folder present if using the folder build?
- Is another copy already running?
- Is port 8000 already in use?

If port 8000 is busy, set another port before launch:

```
$env:BGDDR_PORT = "8001"
.\BGDDR.exe
```

Browser Opens but Page Does Not Load

Try:

- Wait 10-20 seconds after launch.
- Refresh the browser.
- Open `http://127.0.0.1:8000/` manually.
- Check `Logs/bgddr.log`.

Forgot Admin Password

From the app folder:

```
python manage.py changepassword <username>
```

or use another superuser to reset it in Admin -> Users.

PDF Prescription Fails

Use the HTML prescription route and print from the browser. PDF rendering depends on local PDF/native libraries; the clinical workflow and HTML print view can still work.

Lab Result Does Not Update eGFR

Check:

- The test code is `creatinine`.
- The value is numeric.
- The correct unit was selected.
- Date of birth and sex are present on the patient record.

Patient Not in Worklist

Check:

- The follow-up encounter has `next_due_date`.
- The date is not blank.
- You are viewing the correct worklist window.

Study Enrolment Fails

Check:

- Study is not closed.
- Patient meets eligibility rules.
- Trial consent exists if required.
- Patient is not already enrolled in that study.
- RCT arms are configured.

Export Fails

Check:

- You have permission for the requested export.
 - Use de-identified export if you are not a Data Manager.
 - For SPSS export, ensure `pyreadstat` is installed in source mode.
-

26. Quick Reference

Task	Where
Start app	Double-click BGDDR.exe
Login	/login/
Add patient	/patients/add/
Find patient	Patients page or header quick search
Edit patient	Patient page -> Edit
Baseline	Patient page -> Baseline
Follow-up	Patient page -> Follow-up
Lab order	Patient page -> Lab order
Enter lab results	Patient page -> Enter results
Biopsy	Patient page -> Biopsy
Prescription	Patient page -> Prescription
Finalize prescription	Prescription preview -> Finalize
Prescription PDF	/prescriptions/<id>/pdf/
Consent	Patient page -> Consent
Study enrolment	Patient page -> Enrol in study
Adverse event	Patient page -> Adverse event
Worklist	/clinic/worklist/
Analytics	/clinic/analytics/
Export	/clinic/export/
Invite user	/users/invite/
Admin	/admin/
Backup list	python manage.py backup_db --list
Restore list	python manage.py restore_db --list

27. Glossary

Term	Meaning
BGDDR	Bangladesh Glomerular Disease and Diabetes Registry
Encounter	A dated clinical visit
Baseline	Enrolment-time clinical snapshot
LabResult	A dated laboratory value
Prescription	Versioned printable clinical document
Finalized prescription	Immutable prescription that has been reconciled
Treatment exposure	Research medication episode derived from prescriptions or manual entry
Reconciliation	Diffing prescription medicines against open exposure episodes
eGFR	Estimated glomerular filtration rate
UPCR	Urine protein creatinine ratio
UACR	Urine albumin creatinine ratio
UTP	Urine total protein
IFTA	Interstitial fibrosis and tubular atrophy
MEST-C	Oxford IgA nephropathy histologic score
RCT	Randomized controlled trial
SAE	Serious adverse event
CIF	Cumulative incidence function
ICF	Informed consent form

Appendix A. Recommended Clinic Checklist

Before clinic:

- Open BGDDR.
- Check Worklist.
- Confirm recent backup exists.
- Confirm printer/PDF workflow if prescriptions will be printed.

For each patient:

- Confirm identity.
- Open patient record.
- Update demographics if needed.
- Record follow-up.

- Enter available labs.
- Record adverse events or admissions.
- Update consent/study enrolment if relevant.
- Write complete current prescription.
- Review and finalize.
- Print or save prescription.
- Confirm next due date.

After clinic:

- Review unfinalized prescriptions.
 - Review missing next due dates.
 - Check overdue list for follow-up calls.
 - Allow OneDrive sync to complete before closing laptop or moving computers.
-

Appendix B. Administrator Maintenance Checklist

Weekly:

- Confirm backups are being created.
- Review duplicate patient warnings.
- Review user access and inactive accounts.
- Review unfinalized prescriptions.
- Review missing baseline or diagnosis fields.

Monthly:

- Export de-identified research dataset.
- Review data dictionary changes.
- Review study recruitment.
- Review safety summaries.
- Review audit trail for unusual changes.

Before a study opens:

- Confirm study definition.
 - Confirm arms and ratios for RCTs.
 - Confirm trial consent form version.
 - Test enrolment on a non-production/demo patient if available.
 - Confirm exports contain required fields.
-

Appendix C. Files and Folders

Typical desktop package:

```
BGDDR/  
  BGDDR.exe  
  _internal/  
  db.sqlite3  
  Backups/  
  Exports/  
  Media/  
  Logs/
```

Item	Meaning
BGDDR.exe	Desktop launcher
_internal/	Bundled Python/app files for folder build
db.sqlite3	Main local database
Backups/	Automatic and manual database backups
Exports/	Command-line export outputs
Media/	Uploaded/generated files
Logs/	Application log files

Do not delete or move `db.sqlite3` while the app is running.